General concepts in machine learning: Part 2

TASK 3: Science practice

Scientific paper: *A Global and Local SHAP-Driven Interpretable Framework for Early Hospital Admission Prediction With Machine Learning at Emergency Department Triage*
https://doi.org/10.7759/cureus.111163

TASK 3A

Read the **abstract** and identify:

What is the outcome that the authors want to predict?

Ans:_____

What was the prevalence of the event, how often did the event occur in training and test data?

Ans:_____

How many ML prediction models did they try?

Ans:_____

Which models “won”?

Ans:_____

Do a quick search, what type of ML model from our course are CatBoost and XGBoost related to?

Ans:_____

By which metric are CatBoost and XGBoost compared in the abstract?

Ans:_____

Do a quick search, what is a Brier score and how would you calculate it?

Ans:_____

TASK 3B

From the section on Materials and Methods:

How many features did the authors include for the ML models?

Ans:_____

How did they split data into training and test sets?

And how did they split the training data into training and validation sets?

Answer:

The authors considers probabilistic prediction (the models output a probability). What metric was used to define a prediction threshold? Write out the formula for this metric using your notes from PART1 (Wednesday) or a quick search.

Answer:

TASK 3C

What is presented in Table 1?

Can you identify 3 features that you believe will be important for predicting hospital admission?

Answer:

Using Table 2 and the first 5 columns, is CatBoost performing best with respect to all metrics?

In your opinion, is sensitivity or specificity more important for this prediction task? Discuss what these metrics mean in this setting.

Answer:

From Figure 1, what models perform the worst? Why could that be?

Answer:

TASK 3D

Look at and discuss Figure 2, showing global feature importance. Explain how the three ESI levels contributed to predictions. Explain how a persons age contributed to the prediction.

Answer:

Consider the local feature importance plots presented in Figures 4 and 5. What was the predicted probability of hospital admission for these two patients?

Ans:_____

What where the main features contributing to these predictions? How do these relate to the ranking in the global feature importance plot (Figure 3)?

Answer: